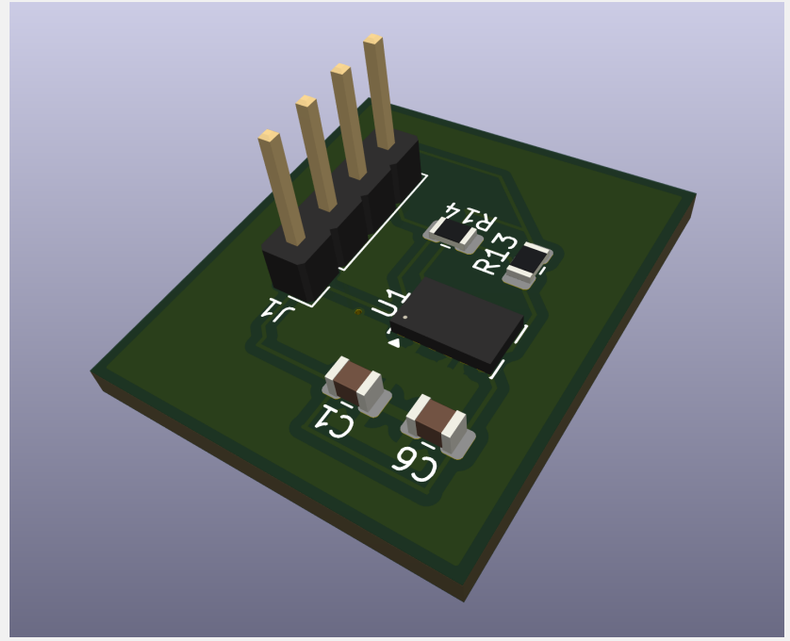
Samana Hassan

ELECTRICAL ENGINEERING

UT Dallas / Expected graduation 2029

AIRCRAFT / ELECTRONICS / CODE



IDEAS INTO HARDWARE.

I’m interested in automotive electronics and PCB design. I like working through design tradeoffs, revisiting calculations, and building alongside people I can learn from.

DESIGN + HARDWARE

MATLAB / KiCad / XFLR5 / SOLIDWORKS / Onshape
Soldering / Wiring / ESC telemetry

CODE + DEVELOPMENT

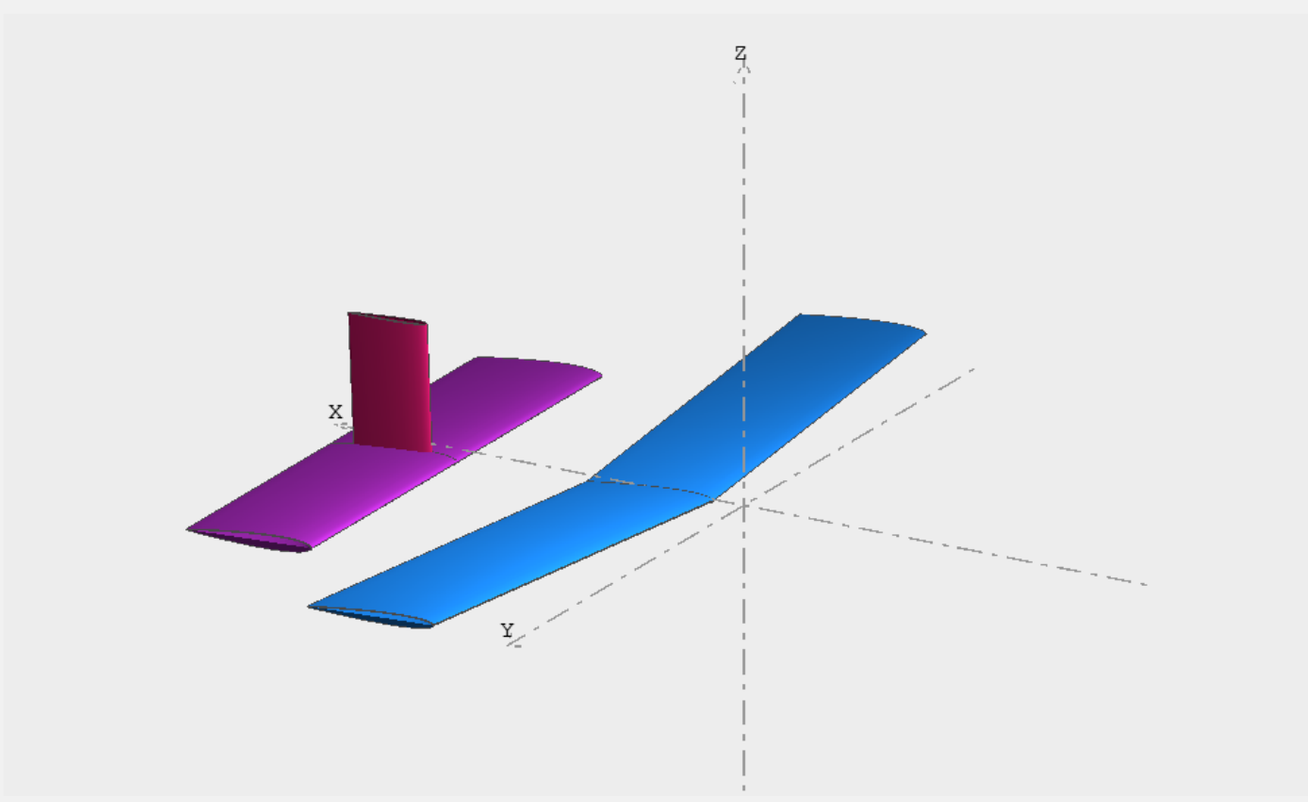
Python / C / Verilog / HTML / CSS / JavaScript
GitHub / VS Code / Quartus Prime / ModelSim

samana7860@gmail.com

Seeking a Summer 2027 internship

01 / AIRCRAFT DESIGN

Speed. Stability. Practical tradeoffs.



XFLR5 aircraft configuration / Three-person team / 2025-26

MY CONTRIBUTION

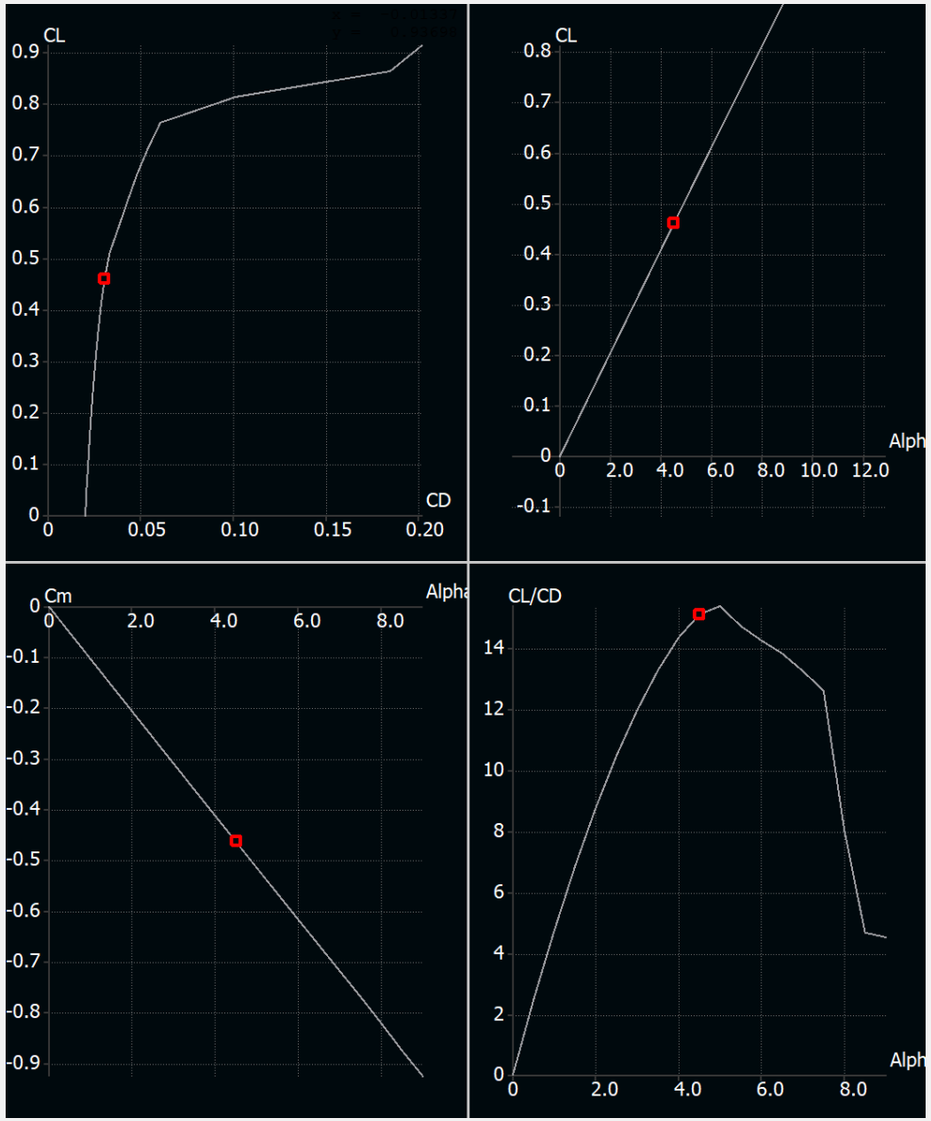
The weekend before preliminary design review, I redid our XFLR5 calculations after learning the analysis process. I used the results to revisit the design with speed and stability in mind and contributed to the preliminary design report.

THE DESIGN DECISION

I initially favored a tapered or elliptical wing. Our team chose a rectangular wing, which was more practical to manufacture.

44 mph

Speed achieved by the team’s aircraft



XFLR5 analysis used to support aircraft design decisions.

THE ANALYSIS BEHIND THE AIRCRAFT

A stable design matters when plans change.

FLIGHT OBSERVATION

After the motor failed, the aircraft was able to glide down from about 100 meters. Seeing the plane remain stable made the analysis feel much more tangible.

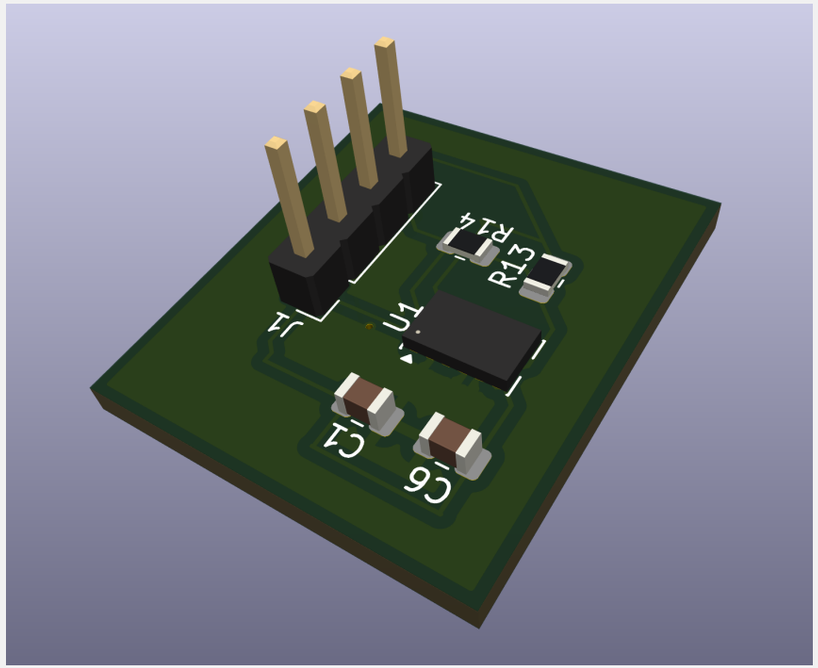
WHAT I LEARNED

A good design has to balance performance with what the team can manufacture. Revisiting the calculations helped me think more carefully about both speed and stability.

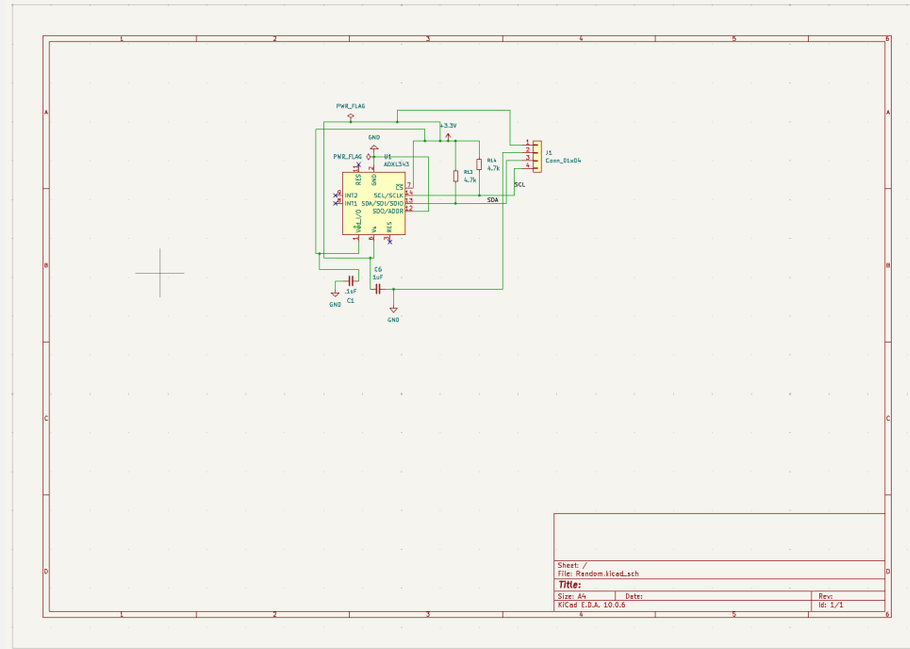
02 / ELECTRONICS

From schematic to board.

A motion-sensing PCB built around an accelerometer / KiCad



3D view of the sensor PCB.



Sensor schematic in KiCad.

MY CONTRIBUTION

I developed the sensor schematic and PCB layout in KiCad and reviewed the board in 3D.

WHAT I LEARNED

The project helped me connect schematic nets with component placement and routing, while exploring my interest in PCB design.

03 / MOTOR PERFORMANCE ANALYSIS

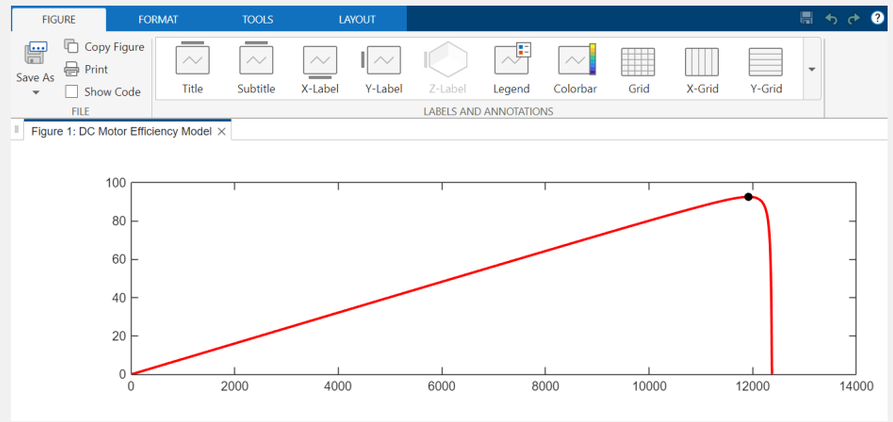
Finding the right power.

```
78   %% ---------------- PLOTS ----------------
79   figure('Name','DC Motor Efficiency Model','Color','w');
80
81   subplot(2,1,1);
82   plot(RPM, eta*100, 'r', 'LineWidth', 2); hold on;
83   plot(RPM_at_max, eta_max*100, 'ko', 'MarkerFaceColor','k');
84   xline(RPM_mark, '--k');
85   plot(RPM(idx_mark), eta_at_mark*100, 'bs', 'MarkerFaceColor','b');
86   text(RPM_mark, eta_at_mark*100, sprintf('  %d RPM', RPM_mark), ...
87       'VerticalAlignment','bottom');
88   xlabel('RPM'); ylabel('Efficiency \eta (%)');
89   title('Motor Efficiency vs Speed (\eta = (V \cdot I - I^2R - P(\omega)) / (V \cdot I))');
90   grid on; hold off;
91
92   subplot(2,1,2);
93   plot(RPM, P_in, 'b', 'LineWidth', 1.5); hold on;
94   plot(RPM, P_cu, 'm', 'LineWidth', 1.5);
95   plot(RPM, P_core, 'g', 'LineWidth', 1.5);
96   plot(RPM, P_out, 'r', 'LineWidth', 1.5);
97   xlabel('RPM'); ylabel('Power (W)');
98   legend('P_{in} = V \cdot I','P_{cu} = I^2R','P_{core}(\omega) [iron+friction]', ...
99       'P_{out} = T \cdot \omega','Location','best');
100  title('Power Balance vs Speed');
101  grid on; hold off;
102
103  %% ---------------- NOTES ----------------
104  % - R, Ke/Kt, kc, kf should be replaced with values from your motor's
105  %   datasheet or from bench measurements (locked-rotor test for R,
106  %   no-load test for Ke and core/friction losses).
107  % - efficiency rises from 0, peaks at a mid-range speed, then falls off as core+friction losses
108  %   grow faster than copper losses shrink, with a marker at 8,000 RPM.
```

Command Window

```
No-load speed        : 12388 RPM
Peak efficiency      : 92.61% at 11917 RPM
Efficiency at 8000 RPM  : 64.35%
Current at 8000 RPM     : 18.89 A
6       if isstruct(pvpairs) && isa(obj,'matlab.graphics.chart.internal.ChartBaseProxy')
K>> ✨ Press Ctrl + Shift + P to generate code with Copilot
```

MATLAB motor analysis code and output.



Plot from the team's motor analysis work.

MY CONTRIBUTION

On the seven-person DBF electronics team, I work with my group on MATLAB code for motor statistics, calculations, and motor-propeller comparisons. We use live ESC telemetry and GitHub to evaluate performance and collaborate on code.

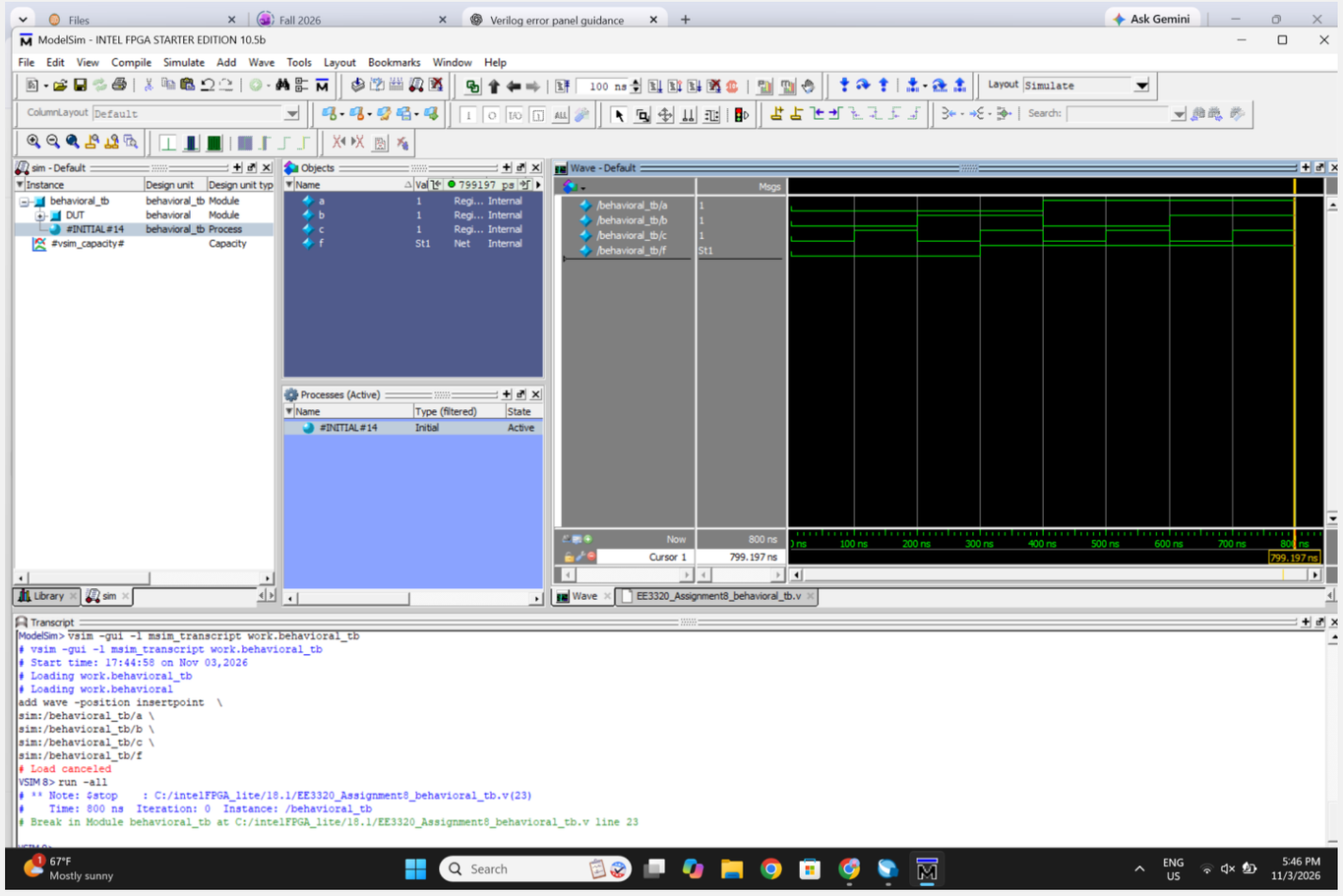
WHY IT MATTERS

The calculations help us compare propulsion options and support component selection.

04 / VERILOG + MODELSIM

One function. Three approaches.

f = a + bc / Behavioral, structural, and dataflow modeling



Simulated inputs and output waveforms in ModelSim.

MY CONTRIBUTION

I modeled the logic function using three Verilog approaches and generated waveforms for specific input combinations in Quartus Prime and ModelSim.

WHAT I CHECKED

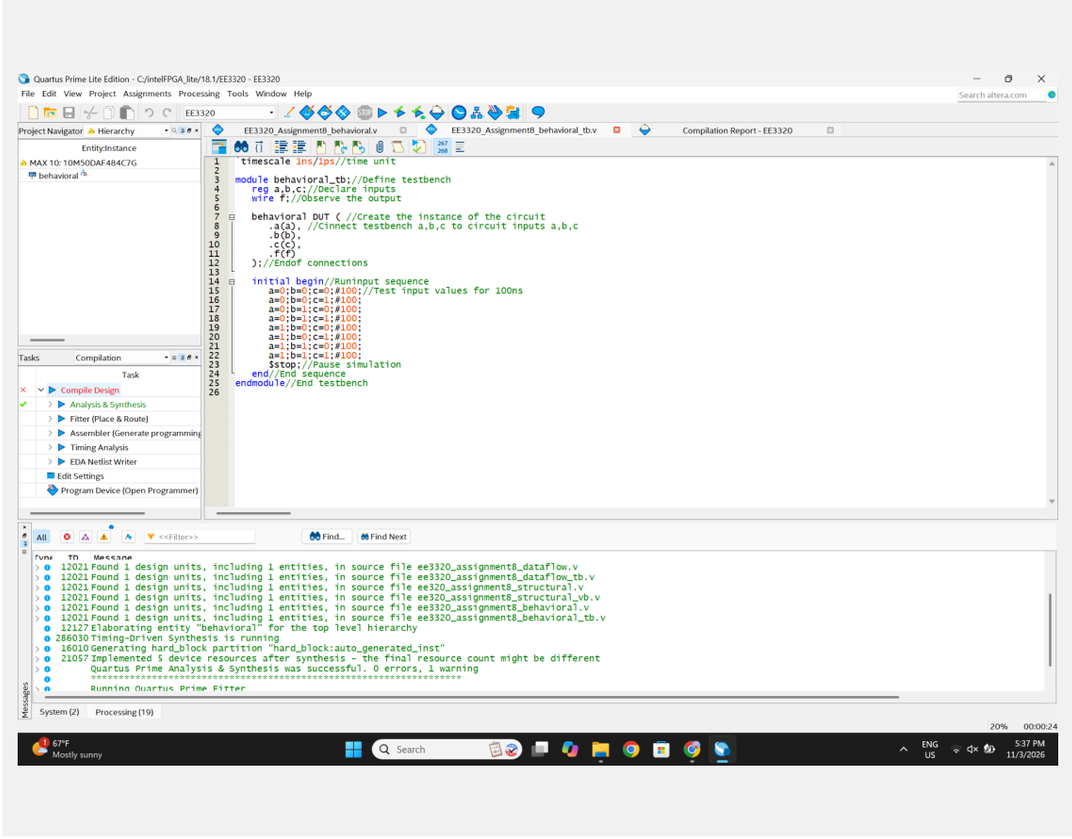
I compared the simulated output with the expected logic: f is 1 when a is 1, or when both b and c are 1.

f = a OR (b AND c)

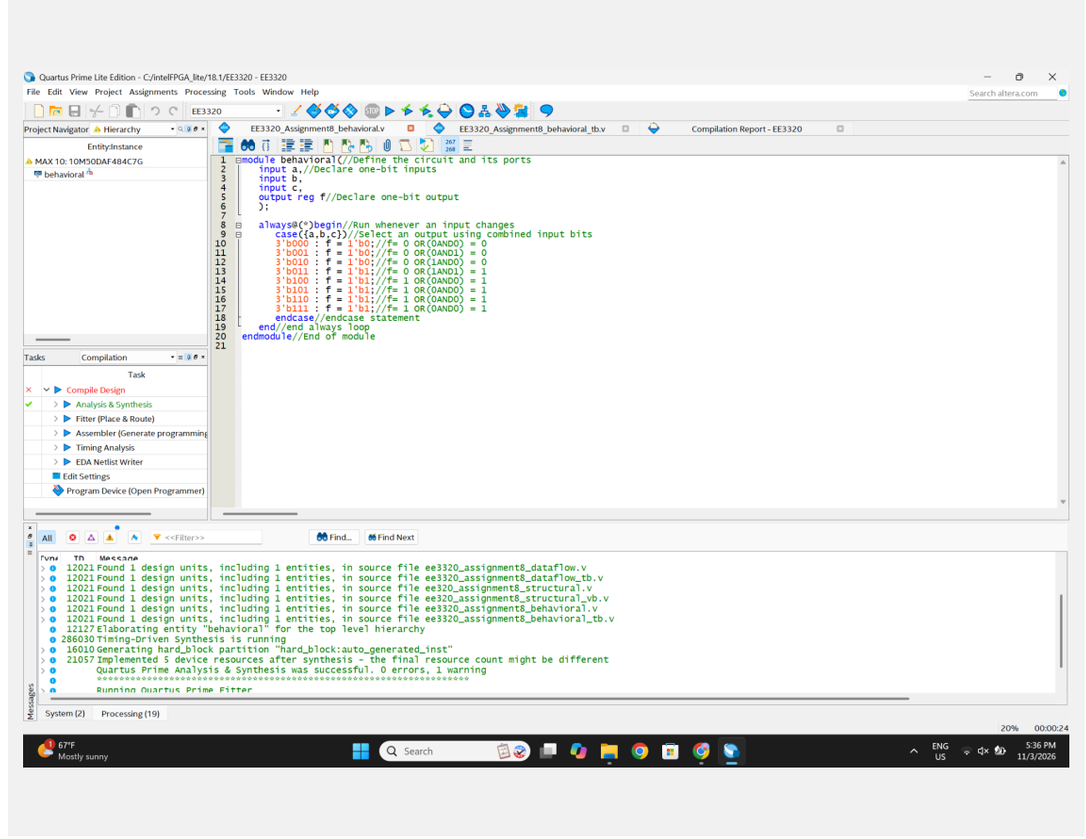
IMPLEMENTATION NOTES

The code behind the waveforms.

Working with multiple modeling styles helped connect the Boolean expression to circuit behavior.



Verilog implementation and simulation work.

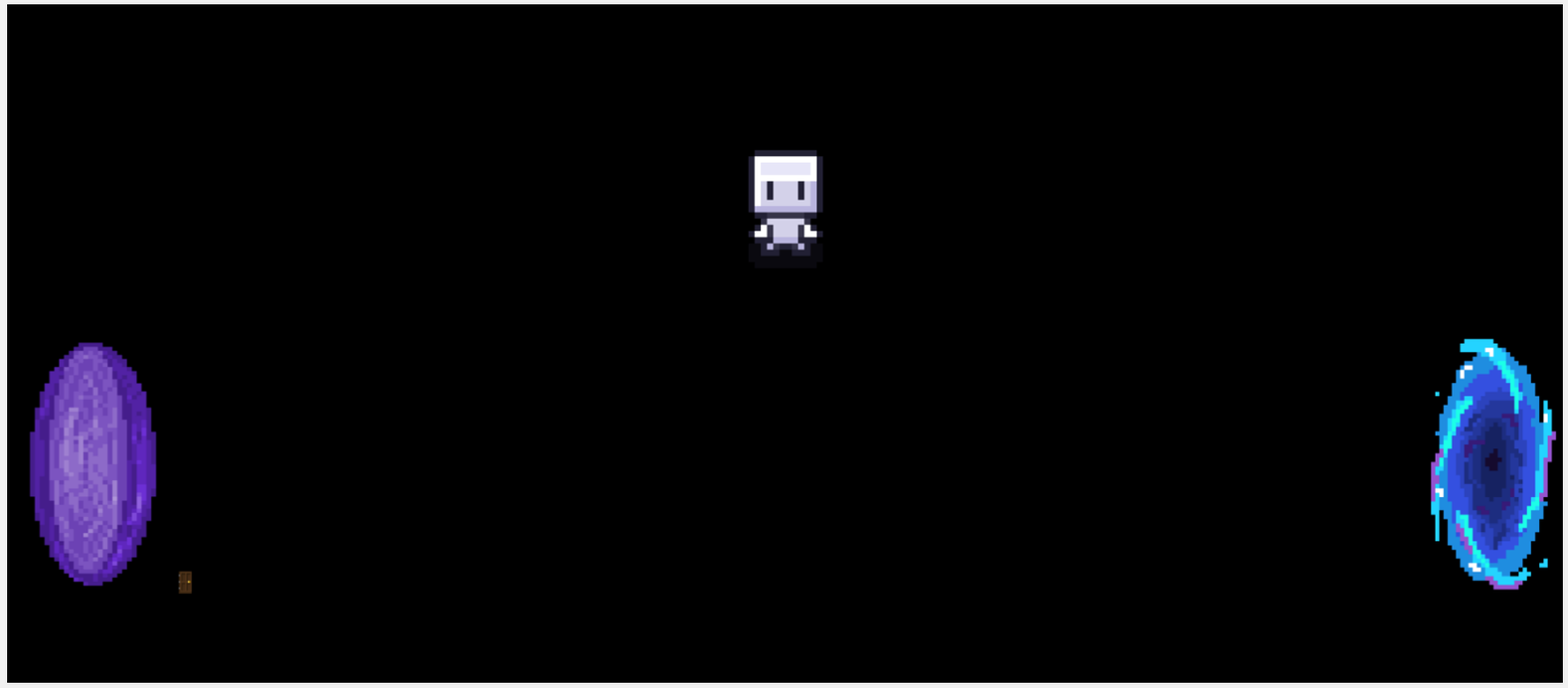


Additional modeling work in Quartus Prime.

Tools: Quartus Prime / ModelSim | Language: Verilog | Context: Digital Circuits coursework

05 / INTERNSHIP PROJECT

A little adventure in the browser.



A scene from my browser-based story game / HTML, CSS, JavaScript

BUILDING + SIMPLIFYING

During my internship, I built websites, including this story game. As the idea grew, the implementation became difficult to manage. I kept working through it and asked peers for help finding a simpler approach.

WHAT I LEARNED

Simplifying the design preserved the key features and made the project more manageable. It taught me to ask for feedback and recognize when a simpler solution better serves the original idea.